# AI Translator Guide

Converting raw country livestock data into the input template

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## Contents

**1 What the AI translator is for** **1**

**2 What the translator will ask, and what it works out for itself** **2**

**3 One inventory per conversation** **2**

**4 The three-step workflow** **2**
4.1 Step 1: Explore . . . 3
4.2 Step 2: Clarify . . . 3
4.3 Step 3: Emit . . . 3

**5 Getting access** **4**
5.1 Initial request (first sign-in) . . . 4
5.2 Approval (administrator side) . . . 4
5.3 Subsequent sign-in (after approval) . . . 4

**6 When to use the translator and when to skip it** **4**

**7 Quality control and limitations** **5**

**8 Privacy and operational cost** **5**
8.1 Related material . . . 5

This page describes what the in-app AI translator does, how to use it for a single inventory, and how access is granted to non-CGIAR email addresses. It complements the user guide and the methodology document with a step-by-step walk-through of the translator workflow.

## 1 What the AI translator is for

The Cattle Uncertainty App expects its data in a specific Excel template: three sheets (Inventory_Metadata, Parameters, Manure_Management), with controlled vocabularies for sub-categories and manure-management systems, and a fixed column order. Real national inventory data rarely arrives in that shape: column names differ, units may be in litres instead of kilograms, sub-category labels reflect local practice ("Local breed adult cows", "draught oxen"), and uncertainty bounds are sometimes reported as a mean ± standard deviation rather than as an explicit 95% confidence interval.

The AI translator converts raw inventory data, in whatever shape it arrives, into the exact template the tool needs. The user uploads an Excel or CSV file (multi-sheet is supported), answers a small number of clarification questions in natural language, and downloads a filled template ready to upload to the Data Input tab.

The translator does column mapping and unit conversion. It does not do the emissions calculation itself; that happens after the filled template is uploaded.

## 2 What the translator will ask, and what it works out for itself

There is nothing to prepare and no form to fill in. The user uploads the file and answers questions as they arrive; the translator asks for what it needs, one or two questions at a time. It deliberately asks after it has read the file rather than before, so that it can ask about the actual column names and sub-category labels in the upload instead of hypothetical ones.

Almost everything comes from the file itself: the sub-category names, the head counts, which parameters are present and which are missing, and the units in use. None of that needs stating in advance.

Four things do not appear in a data file, so the translator will ask about them. They are listed here because they are decisions rather than lookups, and two of them are easy to answer badly under time pressure:

- **Which IPCC edition the inventory reports under**, the 2006 Guidelines or the 2019 Refinement. A reporting decision rather than a property of the data. It determines which manure management systems are available and which default values apply.
- **Where the uncertainty estimates come from**: none, in which case the IPCC catalogue ranges are used; a single expert judgement percentage per value; measured confidence intervals or standard deviations; or a mixture, specified per parameter. This determines whether the bounds in the finished template are the user's own or the catalogue's.
- **How manure is split between management systems**, as a percentage per sub-category summing to 100. Manure management system usage is usually surveyed separately from the herd, so these shares are often absent from the uploaded file and have to be given in the chat.
- **Whether that split is itself uncertain.** If the shares are an estimate rather than a measurement, a lower and an upper percentage can be given for each, and the tool samples within those ranges and renormalises every iteration so the shares still sum to 100. Left unspecified, the split is treated as exact. This one is worth raising unprompted, because nothing in the file will suggest it and an allocation that is really an estimate is otherwise reported as though it were known precisely.

The country name and the inventory year are also required, and are quick to state if they are not in the file.

## 3 One inventory per conversation

Each finished template describes one country and one inventory year, because the Inventory_Metadata sheet carries a single country field. Two inventories therefore need two templates, two conversations and two runs of the tool.

Start a new conversation for the second inventory rather than continuing the first. A conversation carries the sub-category vocabulary, unit conventions and manure system mapping agreed for the inventory already in it, and those are exactly the things that can be carried across to the next one without anyone noticing: the resulting template is still well formed, and the error is only visible by comparing it against the source file.

Two people working on different inventories at the same time should also expect the panel to feel slower than it does for one person alone, because the app serves everyone from a single process. The translation itself is unaffected, but replies queue behind each other. Where several inventories are being prepared in one session, it is worth staggering the work rather than starting every translation at once.

## 4 The three-step workflow

The translator works in three explicit passes: Explore, Clarify, Emit, rather than attempting all three at once. This separation prevents the AI from silently dropping rows or substituting catalogue defaults for values the user supplied.

## 4.1 Step 1: Explore

After the file is read, the panel lists its sheets with their size and lets the user untick reference or lookup sheets the AI does not need before sending; fewer sheets mean a faster first reply and a cheaper run. The AI's first response is then a structured exploration report with four labelled sections:

- **A. File shape.** For each sheet in the upload, the AI classifies the layout pattern (column-oriented, wide-stacked, parameter-labelled, reference-table, or calc-sheet).
- **B. Inventory of values found.** A table listing every (parameter, sub-category) pair the AI could identify in the file, with the source sheet, row, column, central value, lower and upper bounds where present, units, and any qualifier (such as "Local breed only" or "Cross breed weighted average").
- **C. Inventory of gaps.** IPCC catalogue parameters that the file does not provide. These will need to be filled with IPCC defaults at emission time.
- **D. Ambiguities to resolve.** Questions the user must answer before the template can be produced: sub-category vocabulary mapping, unit conversions, biological zeros, manure-management code interpretation, breed disaggregation, and similar.

The exploration report is the persistent ground-truth artefact that drives the rest of the workflow. The user is expected to read it carefully: section B in particular should list every parameter present in the source file. A parameter missing from section B should be raised in the chat before proceeding.

## 4.2 Step 2: Clarify

The user answers the questions in section D in plain natural-language chat messages. There is no required template for the answers; the AI parses them and updates its internal mapping. Typical clarifications include:

- Sub-category vocabulary mapping (e.g., "Cows" → `other_cows`; "Adult males" → `bulls`; "Calves" → split equally into `calves_male` and `calves_female`).
- Unit conversions (litres of milk to kilograms using the cow-milk density 1.03 kg/L; digestibility expressed as a fraction 0.55 versus a percentage 55%).
- Biological zeros (Milk = 0 for non-lactating sub-categories; hours of work = 0 for non-oxen).
- Manure-management code interpretation (e.g., a file-level "PIT" entry → `liquid_slurry`; "DL" → `dry_lot`).
- Whether the file's mean ± SD reporting should be converted to mean ± 1.96 · SD for the 95% confidence interval.

If the user wants to add data that is not in the file (typically national population numbers), the values can be typed directly into the chat. They will be incorporated into the final template alongside the file-derived values.

## 4.3 Step 3: Emit

When the section D questions are answered, the user clicks **Produce template now**. The AI translates section B (file-derived values), section C (catalogue gaps to be filled with IPCC defaults), and the biological zeros confirmed in section D, mechanically, into the JSON template. The workbook is then built once and a green **Download template (.xlsx)** button appears as the only control, with two small links to revise or start over.

Above the button sits a **quality preview**: the same checks the QA/QC tab will run after upload (pass, warn and fail counts with the failing rows) plus a set of consistency notes that compare rows with each other. The notes name the places where a translation typically goes wrong: male and female calves that should carry the same numbers but do not, manure system shares that do not sum to 100 %, one manure system given different coefficients in different production systems, an uncertainty of 0.5 % where the same parameter runs at 50 % elsewhere, and a distribution that cannot represent the value it was given. Each note is a sentence to check against the source file. The user can ask the AI to revise, or download and deal with the failures on the QA/QC tab, where each one can be fixed or explicitly ignored.

The downloaded file follows the official template format exactly. After download, the user should open the file and skim the `data_source` column of the Parameters sheet to confirm that values from the source file are tagged `user_file`, IPCC catalogue values are tagged `ipcc_default`, and explicit zero values are tagged `biological_zero` or `placeholder`. This is the recommended spot-check for catching translation failures before the simulation runs.

# 5 Getting access

The translator is gated behind a magic-link sign-in flow. Every address requires one-time manual approval by the administrator, including CGIAR addresses. Each translation is billed to the project, so access is granted by name rather than by domain. Approval is needed once; after that the browser stays signed in. The full sequence is as follows.

## 5.1 Initial request (first sign-in)

1. Open the **Resources tab** of the Cattle Uncertainty App and locate the AI Translator card (the first item, identified by the green left-edge accent).
2. Enter the email address into the sign-in field and click **Send sign-in link**.
3. Within 1–2 minutes, an email arrives from `noreply@cattle-uncertainty.app` with subject "Sign in to the IPCC Cattle GHG Uncertainty tool". The link in it stays valid for 30 days and may be clicked more than once; after the first successful click the browser stays signed in, so the link is only needed once per browser.
4. Click the link. The app opens in a browser tab and recognises that the email is not yet approved. A confirmation message is displayed:

   > Thanks: your request to access the AI translator (your.email@example.org) has been forwarded to the administrator for approval. You'll receive an email once approved.

   At the same moment, the administrator receives a notification email summarising the request (email address, timestamp).

## 5.2 Approval (administrator side)

5. The administrator reviews the request and, if appropriate, adds the email address to the approved-users list and redeploys the app. The administrator typically responds the same working day.
6. Once added, the user automatically receives a welcome email confirming approval. The email contains a direct link back to the app and a brief reminder of the workflow.

## 5.3 Subsequent sign-in (after approval)

7. Return to the AI Translator card on the Resources tab.
8. Enter the same email address again and click **Send sign-in link**. A fresh magic-link email arrives within 1–2 minutes.
9. Click the new link. The app verifies the email is now on the approved list, drops a long-lived sign-in cookie in the browser, and unlocks the chat panel.
10. On subsequent visits from the same browser, the cookie keeps the user signed in. No additional magic-link step is required unless the user clears site data or switches browsers.

# 6 When to use the translator and when to skip it

The AI translator is intended for the common case in which raw inventory data does not match the tool's template. It is most useful when:

- The user's data is in a non-standard shape (multi-sheet workbook with section headers, form-style layout, free-text comments interspersed with values).
- Column names differ from the template (e.g., `cattle_pop` instead of `N`, `live_wt_lb` instead of `BW`).
- Units are inconsistent (mix of kg and lb, fractions and percentages, litres and kilograms of milk).
- IPCC defaults are needed for parameters the user does not measure.

The translator is not required when the user already has a filled template: in that case, upload directly via the Data Input tab. The translator is also not the appropriate tool for converting an emissions-calculation result; it works on input parameters only.

# 7 Quality control and limitations

The translator performs column mapping and unit conversion. It does not validate the underlying values against external reference data, and it cannot detect systematic errors in the source file (e.g., a transcription mistake in the original survey). Two checks are recommended after every translation:

- **Read the quality preview before downloading.** It lists the checks that will fail after upload and the consistency notes described under Step 3. On the first real translation (a 27-group national inventory) every error the AI had introduced appeared in that list.
- **Spot-check the key numbers.** Compare population, body weight, and milk yield in the downloaded template against the corresponding cells in the source file. Any discrepancy indicates a translation error to be raised in the chat.
- **Review the QA/QC tab after upload.** Any value the translator filled from an IPCC default is flagged with `data_source = "ipcc_default"` in the QA tab, allowing the user to review whether the catalogue default is appropriate for the country context. For body weight (`BW`), the QA tab also runs a continental-default deviation check (Annex Tables 10A.1 / 10A.2 / 10A.3) and warns on deviations larger than ±50%.
- **The workbook records its provenance.** The Notes cell of `Inventory_Metadata` carries the app version, a fingerprint of the instructions the AI worked from, a fingerprint of the IPCC defaults table, the model, and the date, so a file that comes back for support can be traced to the defaults it was built with.
- **Resumed conversations are checked against the current defaults.** If the IPCC defaults or the template changed since a saved conversation was held, the panel says so and asks for the file to be uploaded again before a template can be produced; the earlier discussion would otherwise carry the old numbers into the workbook.

# 8 Privacy and operational cost

The translator is hosted by the GMH project on a dedicated AI service account. The user is not billed for usage. The chat does not retain input data after the session ends; only the user's email address and per-call token counts are logged so the administrator can monitor monthly consumption. A monthly global spending cap is in place; if the cap is reached, an explicit "monthly cap reached, contact administrator" message is shown and translation requests pause until the next cycle.

The IPCC inventory data the translator handles is generally non-sensitive (it is destined for UNFCCC reporting, which is a public-record process). For ministries with internal policies restricting upload of pre-publication data to external AI services, the translator should not be used; the template can instead be filled manually using the IPCC defaults provided in the user guide.

## 8.1 Related material

- User Guide, §"AI-assisted data preparation": the same workflow described in the printable user guide, including the non-CGIAR step-by-step.

- Methodology document: full equations, parameter definitions, and references for everything the translator outputs.
- Find out more: Correlations, the companion topic page covering the correlation modes on the Simulate tab.

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